

TECHNICAL — DATA SCIENCE

Credit Risk Analysis

Predictive modeling on 5,000+ loan applications to guide approval decisions and interest-rate pricing · R (dplyr, ggplot2, randomForest, gbm, pROC)

0.88	0.72	\$1.34M	4
ACCEPTANCE AUC (BOOSTING)	DEFAULT AUC (BOOSTING)	TOTAL EXPECTED PROFIT, 1,000 TEST LOANS	MODELS COMPARED

Executive Summary

The approach began with exploratory data analysis, followed by the creation of new features to enhance predictive accuracy. Four supervised learning methods — linear regression, logistic regression, random forest, and gradient boosting — were applied to the training data. Model performance was evaluated using cross-validated AUC for both loan acceptance and default predictions. Boosting achieved the highest AUC, indicating the strongest discriminative ability. The boosting model was then used with a train-test split to calculate profit outcomes and guide approval decisions for 1,000 new loan applications.

Feature Engineering

Four new features were engineered to capture nonlinear risk factors that raw variables missed:

- **Income-to-loan ratio** — affordability; low ratios indicate financial strain and higher default risk.
- **Loan-to-credit-lines** — how large the requested loan is relative to existing credit history.
- **Inquiries-to-delinquencies** — short-term credit-seeking behavior relative to past repayment problems.
- **Delinquencies-to-credit-lines** — past credit problems relative to available credit.

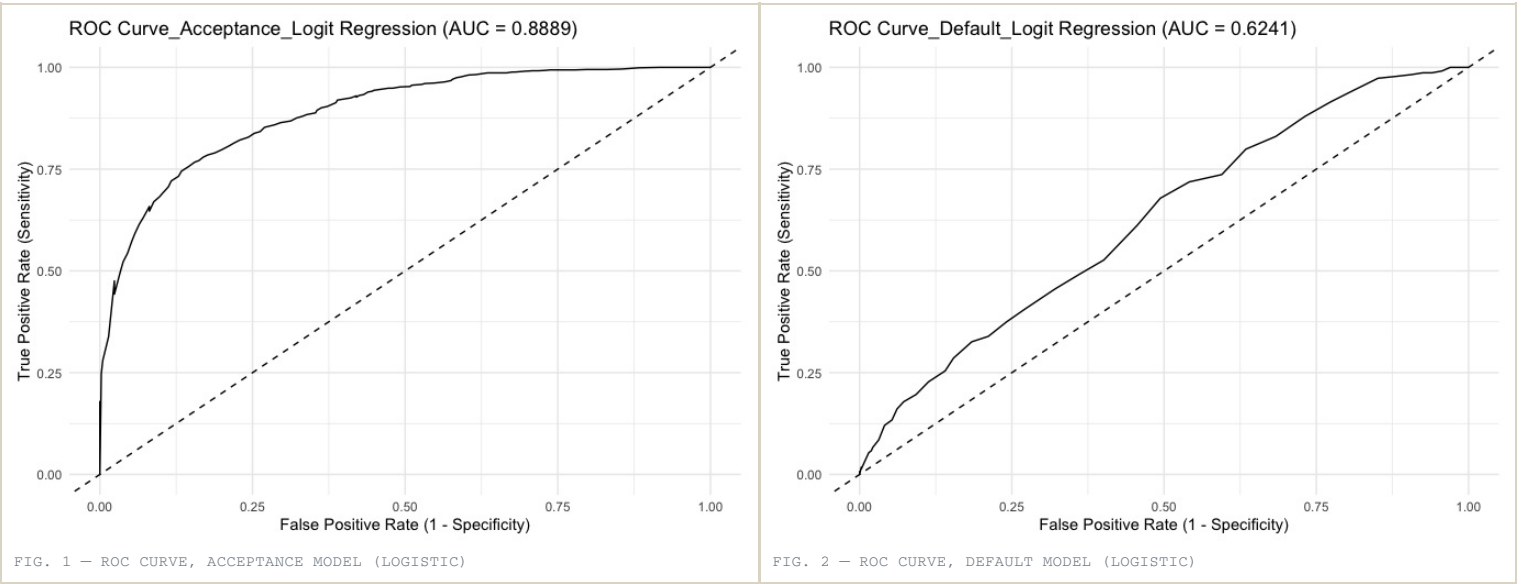
These engineered features reduced noise and improved the model's ability to separate low-risk from high-risk borrowers, particularly for the tree-based algorithms that capture complex interactions.

Model Development & Selection

Four methods were tested and compared by AUC — the most meaningful metric for direct comparison across models:

- **Linear regression** — baseline; performed poorly given nonlinearity in loan risk patterns.
- **Logistic regression** — AUC 0.889 (acceptance) / 0.624 (default); strong interpretability, but struggled with complex feature interactions.
- **Random forest** — AUC 0.812 (acceptance) / 0.555 (default); handled non-linearities well.
- **Gradient boosting (GBM)** — AUC 0.882 (acceptance) / 0.719 (default); outperformed all others, especially on default prediction.

Boosting and logistic regression achieved similar acceptance AUCs, but boosting substantially outperformed on default prediction (~72% vs. ~62%), so boosting was selected as the final model for profit estimation and decision-making.



Business Strategy

Interest rate setting. For each loan, the model evaluated a set of possible interest rates and calculated the expected profit at each one,

selecting the rate that maximized expected profit for that borrower.

Approve / deny decisions. If the optimal expected profit was positive, the loan was approved; if negative, it was denied and profit set to zero — ensuring every issued loan is estimated to add value to the portfolio. Across all 1,000 test loans, this policy produced a total expected profit of **\$1,335,189**.

Risk management. The predicted probability of default was integrated directly into the profit calculation, weighing potential earnings against potential losses for each borrower and identifying the rate that maximizes total profit — without relying on arbitrary thresholds.

Model Interpretation

For **loan acceptance**, the strongest predictors were the offered interest rate, income-to-loan ratio, and annual income — borrowers with higher income and lower income-to-loan ratios were far more likely to accept an offer, while higher offered rates reduced acceptance, showing clear price sensitivity.

For **loan default**, credit utilization, income-to-loan ratio, and credit score were the top predictors — borrowers using a large share of available credit or with lower credit scores were more likely to default.

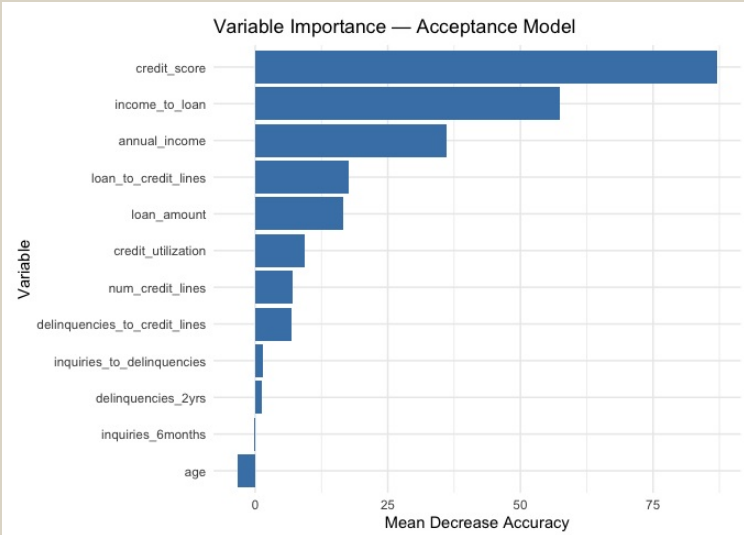


FIG. 3 — VARIABLE IMPORTANCE, ACCEPTANCE MODEL (RANDOM FOREST)

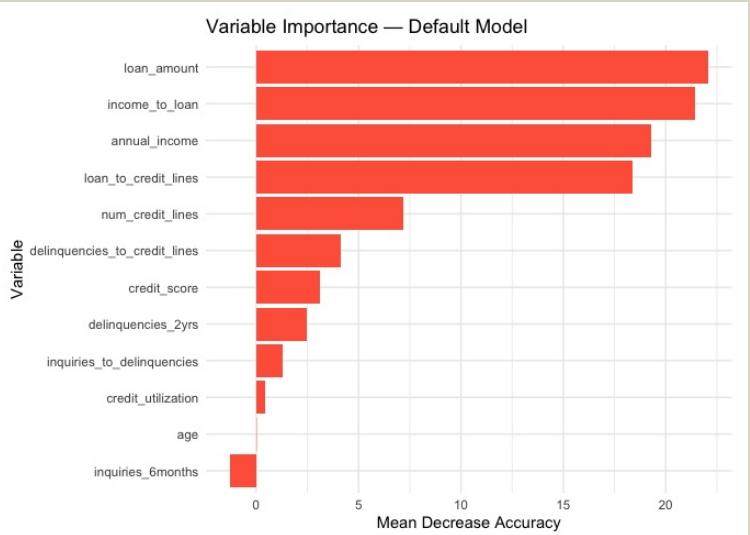


FIG. 4 — VARIABLE IMPORTANCE, DEFAULT MODEL (RANDOM FOREST)

Conclusion

This project combined financial reasoning with machine learning to improve credit decision-making, predicting both who is likely to accept a loan offer and who is likely to default with strong accuracy. The boosting model was the most consistent performer across both tasks. By pairing engineered features with a profit-based decision metric, the resulting system tailors interest rates to each borrower, approves only profitable loans, and manages risk more systematically — a framework that can scale and adapt to new data over time.